

Complex Number 4

- The complex numbers z_1, z_2 and z_3 are represented in the complex plane by the points P, Q and R respectively. If the line segments PQ and PR have the same length and are perpendicular to one another, prove that: $2z_1^2 + z_2^2 + z_3^2 = 2z_1(z_2 + z_3)$
- If w is a complex root of the equation $x^3 = 1$.
 - Show that the other complex root is w^2 .
 - Show that $1 + w + w^2 = 0$.
 - Find in its simplest form, the cubic equation whose roots are $3, 2w + w^2$ and $2w^2 + w$.
- The point P in the complex plane represents the number z . Find and describe the locus of P given that $\frac{1}{z} - \frac{1}{\bar{z}} = i$
- The complex number z is represented by the point C in the Argand diagram. Find and describe the locus of C if $\operatorname{Re}\left(\frac{z-6}{z}\right) = 0$
- Let the point H represent z in the complex plane. Draw the loci of H if (c is constant):
 - $\operatorname{Im}(z^2) = 2c^2$
 - $\operatorname{Re}(z^2) = c^2$
- Show that $(z-2)\overline{(z-2)} = 9$ represents a circle in the Argand diagram.
- Let ω be one of the non-real cube roots of 1
 - Show that $1 + \omega + \omega^2 = 0$
 - Hence or otherwise, prove that $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3 = 0$.
- Consider the polynomial equation: $z^5 - i = 0$
 - Find all the roots of $z^5 - i = 0$. You may leave the roots in the form of $cis \theta$.
 - Hence show that $(z-i)\left[z^2 - \left(2i \sin \frac{\pi}{10}\right)z - 1\right]\left[z^2 + \left(2i \sin \frac{3\pi}{10}\right)z - 1\right] = 0$
 - Hence or otherwise deduce that $\sin \frac{3\pi}{10} - \sin \frac{\pi}{10} = \frac{1}{2}$
- If $\frac{z-1+i}{z+1-i} = ki$, where $k \in \mathbb{R}$. By drawing a diagram or otherwise, find the value of $|z|$. Show all reasoning.
- If $w = \frac{1+z}{1-z}$ and $|z| = 1$ where w and z are complex numbers, determine the locus of w .
- Find and describe the locus of points in the Argand diagram which correspond to $z\bar{z} = (\operatorname{Re}(z-1+3i))^2$
- Solve the following quadratic equations
 - $x^2 + x + 1 = 0$
 - $2x^2 - 4x + 3 = 0$
 - $4x^2 - 4(1+2i)x - (3-4i) = 0$
 - $ix^2 - 2(i+1)x + 10 = 0$
- $3 - 2i$ is one root of $x^2 + bx + c = 0$ where b and c is real. Find b and c .
 - $x^2 + 6x + k = 0$ has one root α where $\operatorname{Im}(\alpha) = 2$. If k is real, find both roots of the equation and the value of k .
 - $1 - 2i$ is one root of $x^2 - (3+i)x + k = 0$. Find k and the other root of the equation.
- $z_1 = 4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$. Write down the modulus and principal argument of
 - z_1^3
 - $\frac{1}{z_2}$
 - $\frac{z_1^3}{z_2}$

15. Write down the moduli and arguments of $-\sqrt{3} + i$ and $4 + 4i$. Hence express in modulus/argument form

a. $(-\sqrt{3} + i)(4 + 4i)$

b. $\frac{-\sqrt{3} + i}{4 + 4i}$

16. $z = 1 + \sqrt{3}i$. find the smallest positive integer n and evaluate z^n if:

a. z^n is real

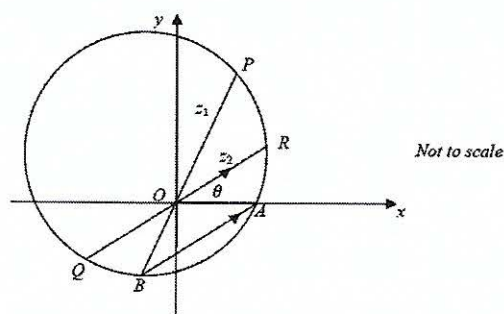
b. z^n is pure imaginary

17. Let ω be the complex root of $z^3 - 1 = 0$ with the least positive argument.

a. Explain why $1 + \omega + \omega^2 = 0$.

b. Find in its simplest form, the polynomial equation with roots $a + b, a\omega + b\omega^2$ and $a\omega^2 + b\omega$ where a and b are real.

18. On the Argand diagram below, point P represents the complex number z_1 and A is the point $(1,0)$. PO is produced to B so that $OB = 1$ unit. A line parallel to AB passing through the origin O meets the circle through points P, A and B at Q and R . Let point R represent the complex number z_2 and $\angle ROA = \theta$.



a. Copy the diagram and prove that $\arg z_1 = 2 \arg z_2$, giving reasons

b. Prove point R represents one of the square roots of z_1 . (You may assume that a cyclic trapezium is isosceles without proof)

Answer:

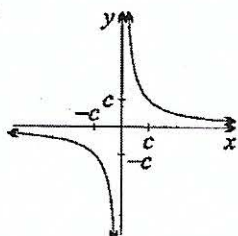
2. $x^3 - 6x - 9 = 0$

3. It is the circle centre $(0, -1)$ with radius 1, omitting the origin.

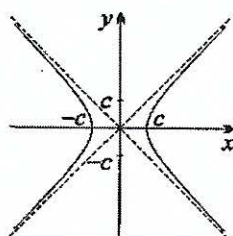
4. It is the circle centre $(3, 0)$, with radius, omitting the origin.

5.

(a)

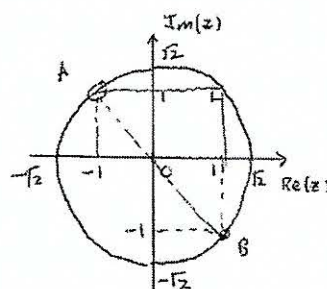


(b)



8. a. $z_1 = \text{cis}\left(-\frac{7\pi}{10}\right), z_2 = \text{cis}\left(-\frac{3\pi}{10}\right), z_3 = \text{cis}\left(\frac{\pi}{10}\right), z_4 = \text{cis}\left(\frac{5\pi}{10}\right) = i, z_5 = \text{cis}\left(\frac{9\pi}{10}\right)$

9. $|z| = \sqrt{2}$, locus of $z: x^2 + y^2 = 2, (z \neq -1 + i)$



10. $w = \frac{y}{1-x}i$, which is purely imaginary. The locus of w is the y -axis (or the imaginary axis).

11. It is a parabola with focus the origin and directrix $x = 1$

12. $= -\frac{1}{2}(1 \pm \sqrt{3}i); x = 1 \pm \frac{1}{\sqrt{2}}i; x = \frac{1}{2} + i, x = -1 - 3i$ or $x = 3 + i$

13. $b = -6, c = 13; -3 \pm 2i, k = 13, k = 8 - i, 2 + 3i$

14. $64, \pi; \frac{1}{2}, -\frac{\pi}{6}; 32, \frac{5\pi}{6}$

15. $2, \frac{5\pi}{6}; 4\sqrt{2}, \frac{\pi}{4}; 8\sqrt{2}\text{cis}\left(-\frac{11\pi}{12}\right); \frac{1}{2\sqrt{2}}\text{cis}\frac{7\pi}{12}$

16. i. $n = 3, z^3 = -8$; ii. No solution ($\because n = \frac{3}{2} + 3k, k$ is integer)

17. b. $x^3 - 3abx - (a^3 + b^3) = 0$